

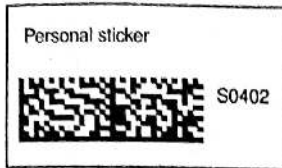


2021-2022 winter semester exam

Machine learning (Technische Universität München)



Scanne, um auf Studocu zu öffnen



Compliance to the code of conduct

I hereby assure that I solve and submit this exam myself under my own name by only using the allowed tools listed below.

Alon Ben-Arleh

Signature or full name if no pen input available

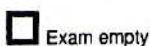
Maschinelles Lernen

Exam: IN2064 / Final exercise
Examiner: Prof. Dr. Stephan Günnemann

Date: Thursday 17th February, 2022
Time: 17:00 – 19:00

Working instructions

- You have to sign the code of conduct. (Typing your name is fine).
- You have to either print this document, solve the problems on paper and then scan your solutions OR paste scans/pictures of your handwritten, on-paper solutions into the solution boxes in this PDF.
- You must not use any other means of creating a submission (e.g. digital pen on a tablet).
- Make sure that the **QR codes are visible** on every uploaded page. Otherwise, we cannot grade your submission.
- **You must solve the specified version of the problem.** Different problems may have different versions: e.g. Problem 1 (Version A), Problem 5 (Version C), etc. If you solve the wrong version you get **zero** points.
- Only write on the provided sheets, **submitting your own additional sheets is not possible.**
- The last pages (after problem 11) can be used as scratch paper.
- All sheets (save for empty scratch paper) have to be submitted to the upload queue. Missing pages will be considered empty.
- **Only use a black or blue color (no red or green)! Pencils are allowed.**
- **For problems that say "Justify your answer" you only get points if you provide a valid explanation.**
- **For problems that say "Derive" you only get points if you provide a valid mathematical derivation.**
- **For problems that say "Prove" you only get points if you provide a valid mathematical proof.**
- If a problem does not say "Justify your answer", "Derive" or "Prove" it's sufficient to only provide the correct answer.
- Instructor announcements and clarifications will be posted **on Piazza** with email notifications.
- Exercise duration - 120 minutes.



Problem 1 (Version C) (10 credits)

0
1
2
3
4
5
6
7
8
9
10

page 4

$$p(x|y) = \frac{p(y|x) \cdot p(x)}{p(y)}$$

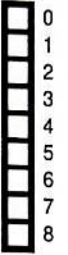
$$p(\theta|D, \lambda, \alpha) = \frac{p(D|\theta, \lambda, \alpha) \cdot p(\theta|\lambda, \alpha)}{p(D|\lambda, \alpha)}$$

$$p(D|\theta, \lambda, \alpha) = \prod_{i=1}^n p(x_i|\theta, \lambda, \alpha) = \prod_{i=1}^n \begin{cases} \frac{1}{\theta} & , 0 \leq x_i \leq \theta \\ 0 & , \text{else} \end{cases}$$

$$\Rightarrow \underline{x_i > \theta}: p(\theta|D, \lambda, \alpha) = 0 \quad \begin{matrix} \text{(because } p(D|\theta, \lambda, \alpha) = 0 \\ \text{because } 0 \cdot (\text{anything}) = 0 \end{matrix}$$

$$\underline{\forall x_i, 0 \leq x_i \leq \theta}: p(D|\theta, \lambda, \alpha) = \frac{1}{\theta^n}$$

$$\cancel{\lambda > \theta} \quad \underline{\lambda > \theta}: p(\theta|D, \lambda, \alpha) = 0 \quad \text{because } p(\theta|\lambda, \alpha) = 0$$



from the lecture: $W^* = (X^T X)^{-1} \cdot X^T \cdot y$

$$\Rightarrow W_{\text{new}}^* = (X_{\text{new}}^T X_{\text{new}})^{-1} \cdot X_{\text{new}}^T \cdot y_{\text{new}}$$

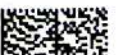
we want to determine: $W^* \stackrel{!}{=} W_{\text{new}}^*$

$$\Rightarrow (X^T X)^{-1} \cdot X^T \cdot y = (X_{\text{new}}^T X_{\text{new}})^{-1} \cdot X_{\text{new}}^T \cdot y_{\text{new}}$$

$$y_{\text{new}} = \sigma \cdot y \Rightarrow (X^T X)^{-1} \cdot X^T \cdot y = (X_{\text{new}}^T X_{\text{new}})^{-1} \cdot X_{\text{new}}^T \cdot \sigma \cdot y$$

$$\Rightarrow \sigma \cdot (X^T X)^{-1} \cdot X^T \cdot y = (X_{\text{new}}^T X_{\text{new}})^{-1} \cdot X_{\text{new}}^T \cdot y$$

$$\Rightarrow \text{let } X_{\text{new}} = \sqrt{\sigma} \cdot X$$



Problem 3 (Version D) (3 credits) - 13

0 1 a)

a- L_1
b- L_2
c- L_∞

0 1 b)

~~L_1 : using $k=3$, $| (8,6) - (6,4) |_1 = 4$
 $| (8,6) - (6,3) |_1 = 5$
 $| (8,6) - (4,6) |_1 = 4$
So we
there are only 2
data points of class 1. As we use $k=3$ nearest
neighbors, that is not well defined what the result will
be. So I think, that as only class 2 has 3
data points, for all norms, L_1, L_2, L_∞~~

0 1 c)

~~the points will be classified as class 2.
If we decide to actually count the error:
 $P(Y=1|X) = \frac{1}{5}$~~

L_1 : Three NN are: $(6,4)$, $(6,3)$, $(4,6)$, $(8,10)$
so out of

(b and c on pg. 16)

Problem 4 (Version B) (6 credits) -15

a)

As the sets have the same shape ~~given~~
then we can assume it is generated by
the same covariance matrix, so we can
choose LDA.

b)

same as a.

c)

we want a ~~q~~ quadratic ^{decision} boundary so
we'll choose Naïve Bayes.



Problem 5 (Version A) (10 credits) -18

a)

$$\max_i (x_i) = \max(x_1, x_2, \dots, x_n)$$

$$\max_i (x_i) = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} (x_1, x_2, \dots, x_n), \text{ where the } 1 \text{ is in the } i \text{ of the max value}$$

$$\min_i (x_i) = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \text{ where the } 1 \text{ is in the } i \text{ of the min value}$$

$\Rightarrow \max_i (x_i) - \min_i (x_i)$ is a linear combination of x_i and therefore it is convex. Another

b)

$$g(x) = \frac{1}{n} \sum_i^n |x_i - \min_{t \in \mathbb{R}} \|x - t\|_1|$$

proof on page 15.

Problem 6 (Version B) (8 credits) 24-25

a)

$$\begin{aligned}
 z &= f(x, y) \\
 t &= g(z) \Rightarrow t = g(f(x, y)) \\
 \Rightarrow \frac{\partial t}{\partial y} &= \frac{\partial g}{\partial z} \cdot \frac{\partial f}{\partial y} \\
 \frac{\partial t}{\partial x} &= \frac{e^x}{e^x} \cdot \frac{\partial g}{\partial z} = e^{(x-y)} \cdot d_{\text{out}} \\
 \Rightarrow f(x, y) &= e^{(x-y)} \Rightarrow \boxed{\text{out} = \text{np.exp}(x-y)}
 \end{aligned}$$

b)

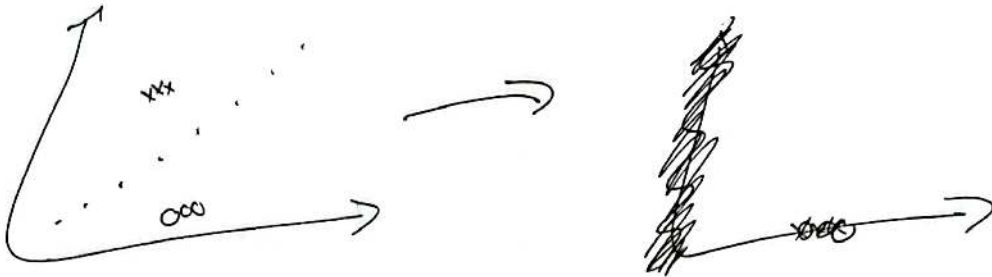
$$\begin{aligned}
 \text{ex } g &= \text{sum}(f(x, y)) = \text{sum}(z) = (1 \ 1 \ 1 \dots 1) \cdot \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \\
 \frac{\partial g}{\partial z} &= \text{ones}(\text{length} = \text{length}(z)) \\
 \boxed{d_z &= \text{np.ones}(\text{length}(z)) \cdot d_{\text{out}}}
 \end{aligned}$$



a)

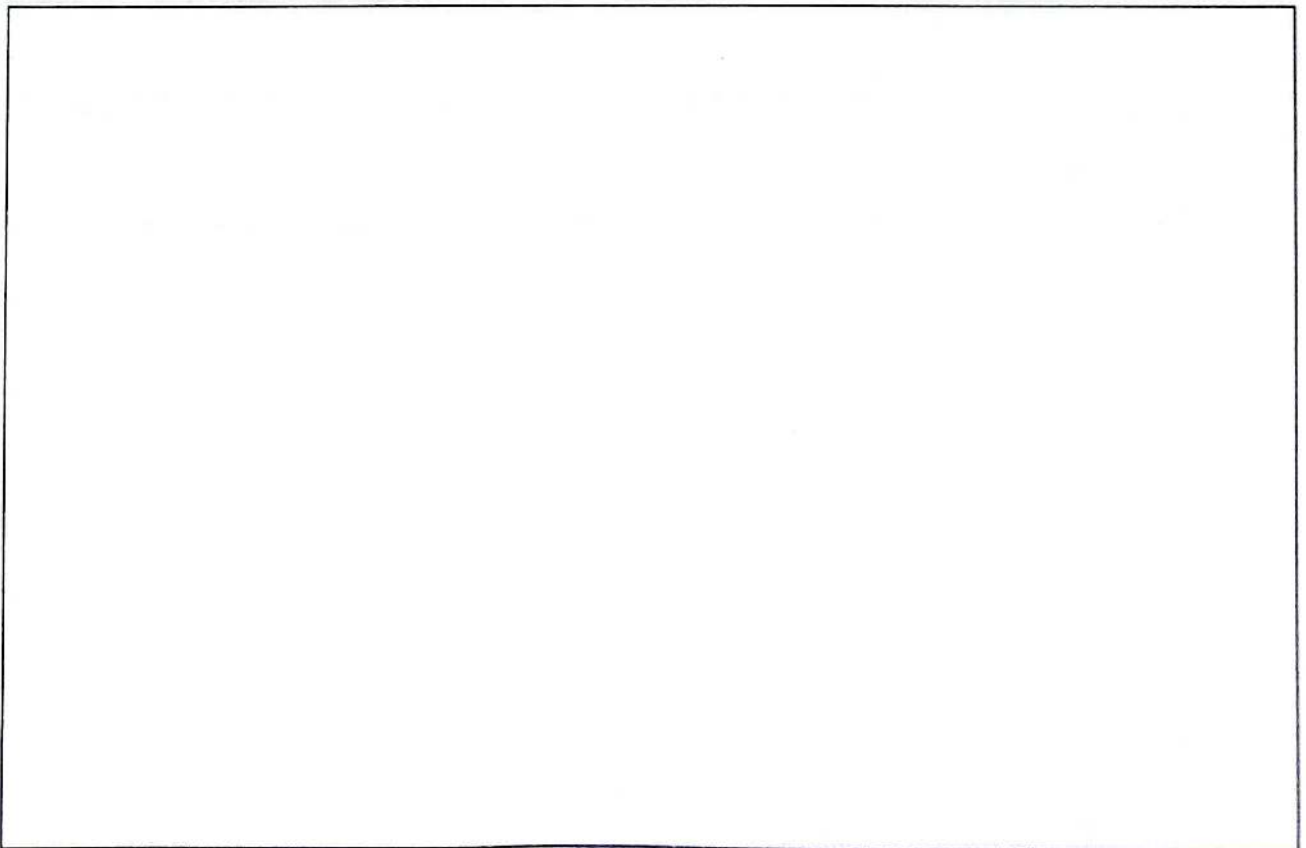
no, it is not, as we lose information ($k < D$).

(2,1)



We reduced from 2D to 1D and we only kept the high-variance dim. so the data is no longer linearly-separable.

b)



A - ~~1450~~ $k=50$

B - $k=5$

C - $k=20$

the higher the k , the less information we lose. ~~so we'd expect~~

On the other hand, the more flexibility the model has, the higher the chance to overfit. So the highest train error but steady test error - $k=5$, Model B.

lowest train error, high test error - Model A $k=50$

And we're left with the intermediate - Model C $k=20$





M-step: Re-estimate the params:

$$\mu_k^{(t-1)}, \pi_k^{(t-1)} \quad \leftarrow \text{uniform } \frac{1}{K}$$

We solve: $E[\log p(x, z | \pi, \mu, \Sigma)] = E[\log \prod_{j=1}^N p(x_j, z_j)]$

$$= E[\log \prod_{j=1}^N \prod_{k=1}^K \frac{1}{K} \exp(-\frac{1}{2\sigma^2} (x_{j,d} - \mu_k)^2)]$$

$$E[\log p(z, x | \theta)] = \sum_{i=1}^N \sum_{k=1}^K \gamma(z_i = k) \cdot \log p(z_i, x_i | \theta)$$

=

Problem 10 (Version D) (10 credits) - 45

a)

$$f(x) = Ax + b$$

$\begin{matrix} & \nearrow & \nearrow & \nearrow \\ 4 \times 4 & & 4 \times 1 & 4 \times 1 \end{matrix}$

$$\Delta_1 = \sup_{x \approx x'} \|f(x) - f(x')\|_1$$

$$Ax + b = \begin{pmatrix} 8 & 9 & 8 & 0 \\ 2 & 7 & 5 & 8 \\ 0 & 8 & 0 & 5 \\ 9 & 7 & 7 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + b$$

$$f(x) - f(x') = Ax + b - (Ax' + b) = A(x - x') = A \cdot e^{(d)}$$

where $e^{(d)}$ is the standard basis vector, with (d) being the index of the different component.

~~$\Rightarrow$ for $e^{(1)}$: $\Delta_1 =$~~

$\Rightarrow \|A \cdot e^{(d)}\|_1 = \text{sum of } (d)^{\text{th}} \text{ column}$

$$\Rightarrow e^{(1)} \Rightarrow 8 + 2 + 0 + 9 = 19$$

$$e^{(2)} \Rightarrow 9 + 7 + 8 + 7 = 31$$

$$e^{(3)} \Rightarrow 8 + 5 + 0 + 7 = 20$$

$$e^{(4)} \Rightarrow 0 + 8 + 5 + 1 = 14$$

$$\Rightarrow \Delta_1 = 32$$

0
1
2
3
4
5
6



0
1

b)

$$\sigma = \frac{\Delta_1}{\epsilon}, \text{ for } \epsilon\text{-DP} \Rightarrow \left[\sigma = \Delta_1 \cdot 2 = 64 \right]$$

0
1
2
3

c)

$$\Delta_\infty = \sup_{x \approx x'} \|f(x) - f(x')\|_\infty$$

We can use the group privacy:

If $\mathcal{M}$ is an ϵ -DP mechanism with respect to $x \approx x'$ then $\mathcal{M}$ is $(t\epsilon)$ -DP w.r.t changes of t instances ~~individually~~ as we have $4D$, $4 \cdot \frac{1}{2} = 2$ and we got a 2-DP Mechanism.



Problem 11 (Version A) (11 credits) - 46

a)

ID	1	2	3	4	5	6
x_1	-4	-1	1	2	1	-3
x_2	2	-5	-3	-5	-1	-2
A	a	a	a	b	b	b
r	0	0	1	0	1	1
r	0	1	0	0	1	1

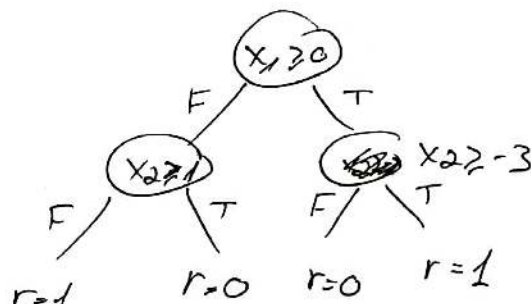
b)

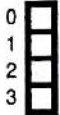
Independence: $P(R=1|A=a) = P(R=1|A=b)$

$$P(R=1|A=a) = \frac{1}{3}$$

$$P(R=1|A=b) = \frac{2}{3}$$

We can either add one to a or remove one from b . We'll ~~remove~~ ^{add} ~~id 3~~ ^{to a} from a ($r=0 \rightarrow 1$) by changing the threshold $x_2 \geq -2 \Rightarrow x_2 \geq -3$





c)

separation: $P(R=1|Y=1, A=a) = P(R=1|Y=1, A=b)$ TP

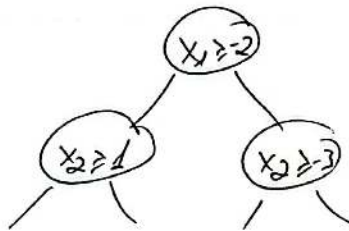
$P(R=1|Y=0, A=a) = P(R=1|Y=0, A=b)$ FP

$TP_a: 0$, $TP_b: 1$

$FP_a: \frac{1}{2}$ $FP_b: 0$

for changing $r_3 = 0 \rightarrow 1$ we can do the same change as in ~~the~~ section 6.1.

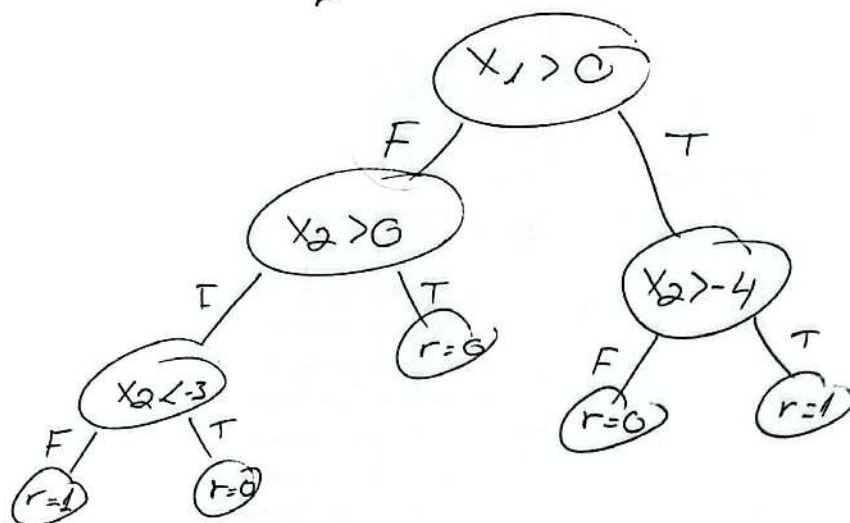
Also, we'll change the root node to effect Id 2 ($r=1 \rightarrow 0$)



d)

Eco:

$$P(R=1|Y=1, a) = P(R=1|Y=1, b) \quad TP$$



I managed to construct a tree that classifies correctly all of the data points.

e)

same as d.



problem 3: (ver. D)

b)

$$|(8,6)-(6,4)|_1 = 4$$

$$|(8,6)-(6,3)|_1 = 5$$

$$|(8,6)-(8,10)|_1 = 4$$

$$|(8,6)-(4,6)|_1 = 4$$

$$|(8,6)-(5,9)|_1 = 6$$

3-NN: 2 class 2

1 class 1

$$\Rightarrow \frac{1}{g} = 2$$

c)

$$|(8,6)-(6,4)|_1 = 2$$

$$|(8,6)-(6,3)|_1 = 3$$

$$|(8,6)-(8,10)|_1 = 4$$

$$|(8,6)-(4,6)|_1 = 4$$

$$|(8,6)-(5,9)|_1 = 3$$

2 class 1

1 class 2

$$\Rightarrow \frac{1}{g} = 1$$



Problem 5 - A:

We know that $\max$ is a convex func above $\mathbb{R}$ also $\min$ is a concave func.

therefore we can say that $(- \min)$ is convex, and the sum of these funcs is convex as a sum of convex funcs.

